

HR EMPLOYEE ATTRITION ANALYSIS REPORT

A data-driven review of workforce retention, attrition risk, employee demographics, compensation, and career-development patterns.

Dataset: HR Employee Attrition

Records analyzed: 1,470 employees

Tools: Excel | SQL Server Management Studio | Power BI | DAX | Power Query

Scope: Employee attrition, workforce composition, compensation, job satisfaction, tenure, promotion, and training patterns

Purpose of the report

This report combines cleaned employee data, SQL analysis, and four interactive Power BI dashboard pages to identify where attrition is concentrated, which workforce segments are most exposed, and which retention questions deserve management attention. The analysis is descriptive: it identifies patterns and associations in the data, not proof of causation.

1,470 Total Employees	1,233 Employees Stayed	237 Employees Left	16.12% Overall Attrition Rate	153 High-Risk Employees
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Executive Summary

The organization has 1,470 employees, of whom 237 left, producing an overall attrition rate of 16.12%. The results point to concentrated risk rather than a uniform retention issue. Sales Representatives have the highest job-role attrition rate at 39.76%, and lower job satisfaction is associated with materially higher attrition. Employees who left earned less on average than employees who stayed, while managers and research directors show substantially longer tenure.

Priority area	Evidence	Management interpretation
Frontline role retention	Sales Representatives: 39.76% attrition; Laboratory Technicians: 23.94%; HR role: 23.08%.	Prioritize role-specific retention reviews, workload checks, onboarding support, and career-path clarity.
Compensation gap	Average monthly income: 4,787.09 for leavers vs 6,832.74 for stayers.	Review pay competitiveness and progression for roles with elevated exits; interpret alongside job level and role mix.
Early-tenure risk	0-1 Years group: 34.88% attrition; 2-3 Years: 18.43%.	Strengthen first-year onboarding, manager support, and early career development.
Satisfaction and overtime	High-risk definition: OverTime = Yes and Job Satisfaction <= 2; 153 employees meet it.	Use targeted listening, workload monitoring, and manager interventions for this segment.
Tenure concentration	Managers average 14.43 years; Sales Representatives average 2.92 years.	Build retention pathways in low-tenure, high-attrition roles and preserve knowledge in long-tenure roles.

Headline Recommendations

- **Focus retention plans on Sales Representatives and Laboratory Technicians.** Their attrition rates are considerably above the overall 16.12% rate.
- **Address early employee experience.** The first year shows the highest tenure-group attrition, which makes onboarding and line-manager support high-value actions.
- **Use compensation and career discussions together.** The income gap between leavers and stayers is notable, but should be assessed by job role and level before policy decisions.
- **Monitor the 153 high-risk employees.** Overtime combined with low job satisfaction is a practical, interpretable screening flag for HR follow-up.

Data Preparation and Analytical Method

The project began with an HR Employee Attrition dataset containing 1,470 employee records. Data preparation was completed before analysis so that Excel, SQL Server, and Power BI would use a consistent, analysis-ready structure.

Step	Result / purpose
Missing-value review	No missing values were identified in the working dataset.
Duplicate review	No exact duplicate employee records were found.
Text standardization	Whitespace was trimmed and label formatting was standardized, including Business Travel labels.
Column standardization	Column names were converted to lowercase snake_case for consistent SQL and Power BI use.
Constant columns removed	employee_count, over18, and standard_hours were removed because they do not vary across employees.
Derived flags	attrition_flag and overtime_flag were created for analysis-ready binary logic; high-risk analysis uses overtime plus low job satisfaction.
Final analysis data	1,470 rows and 34 columns available for Excel, SSMS, and Power BI analysis.

Analytical workflow

1. Excel: department attrition rates, job role by job satisfaction cross-tab, and high-risk employee flag. 2. SQL Server: group-level attrition rates, average-income comparison, and tenure ranking. 3. Power BI: four-page interactive report with slicers, navigation buttons, KPI cards, and cross-filtering visuals.

Important scope note: the dataset does not include exit dates, recruitment cost, performance narrative, manager quality, external labor-market data, or causal experimentation. Findings should therefore inform further investigation and targeted action, not be treated as proof of why any individual employee left.

Power BI Dashboard 1 - Executive Overview

The Overview page provides the main retention story: 1,470 employees, 237 exits, 16.12% attrition, 153 high-risk employees, department comparison, gender distribution among leavers, overtime-related exits, and employee-tenure patterns.

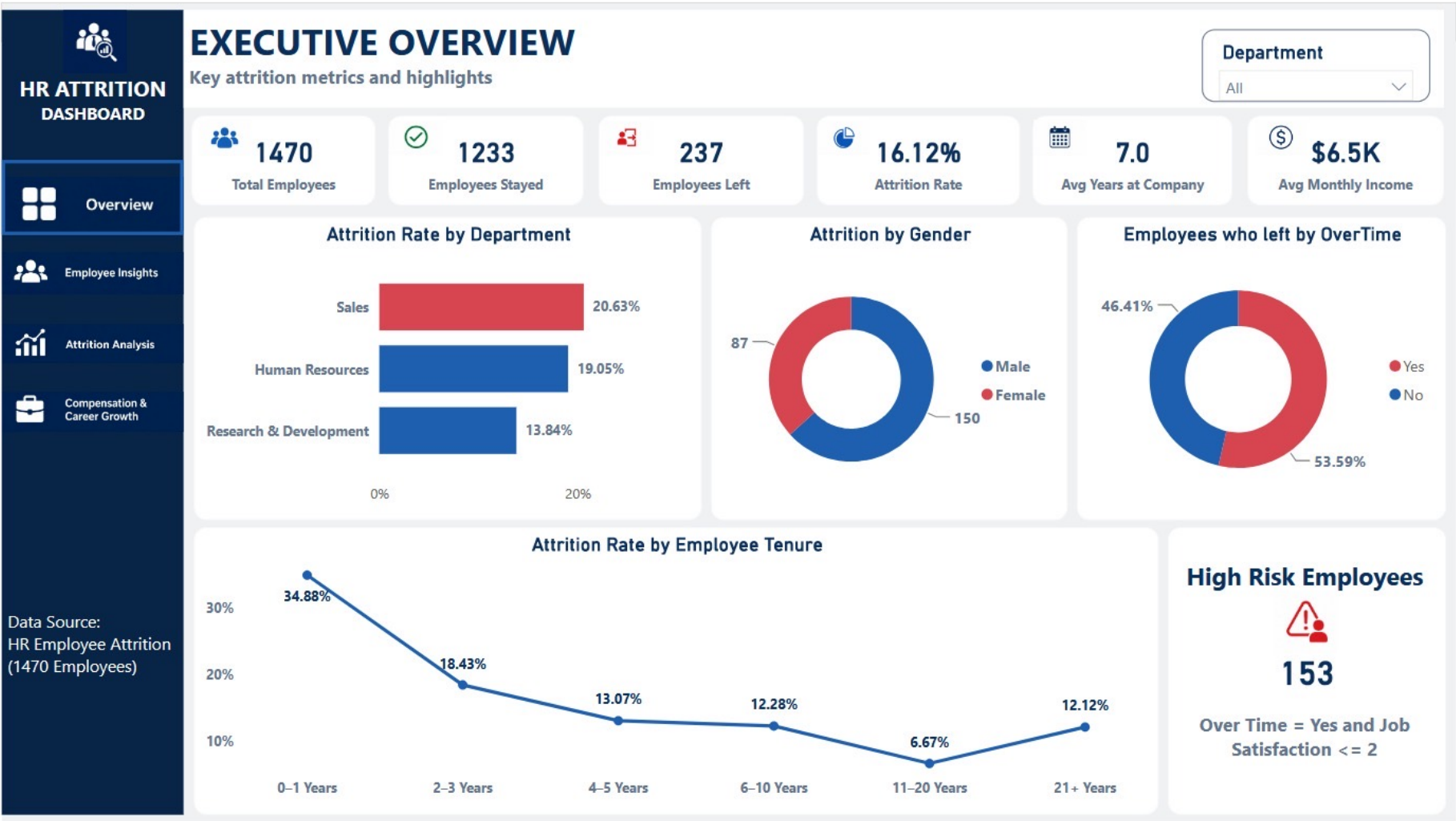


Figure 1. Executive Overview dashboard: workforce size, retention, attrition, department patterns, overtime exits, tenure, and high-risk employees.

Power BI Dashboard 2 - Employee Insights

The Employee Insights page profiles the workforce by age group, education field, marital status, income and tenure by gender, plus job-satisfaction distribution and attrition rate by satisfaction score.

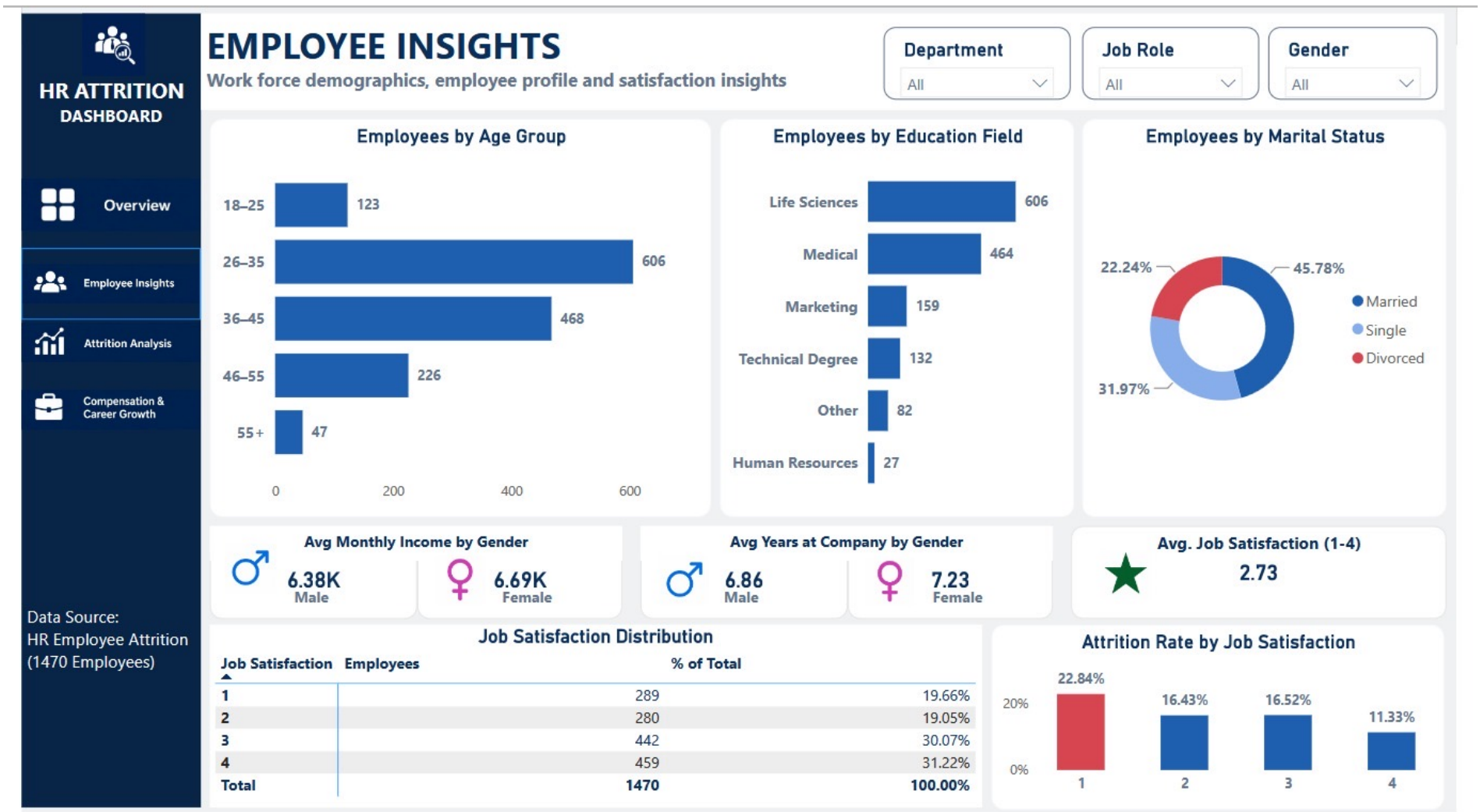


Figure 2. Employee Insights dashboard: workforce composition, demographics, income, tenure, and job satisfaction.

Power BI Dashboard 3 - Attrition Analysis

The Attrition Analysis page isolates risk patterns by job role, job satisfaction, business travel, stock-option level, and performance rating. It supports targeted follow-up rather than a generic retention response.

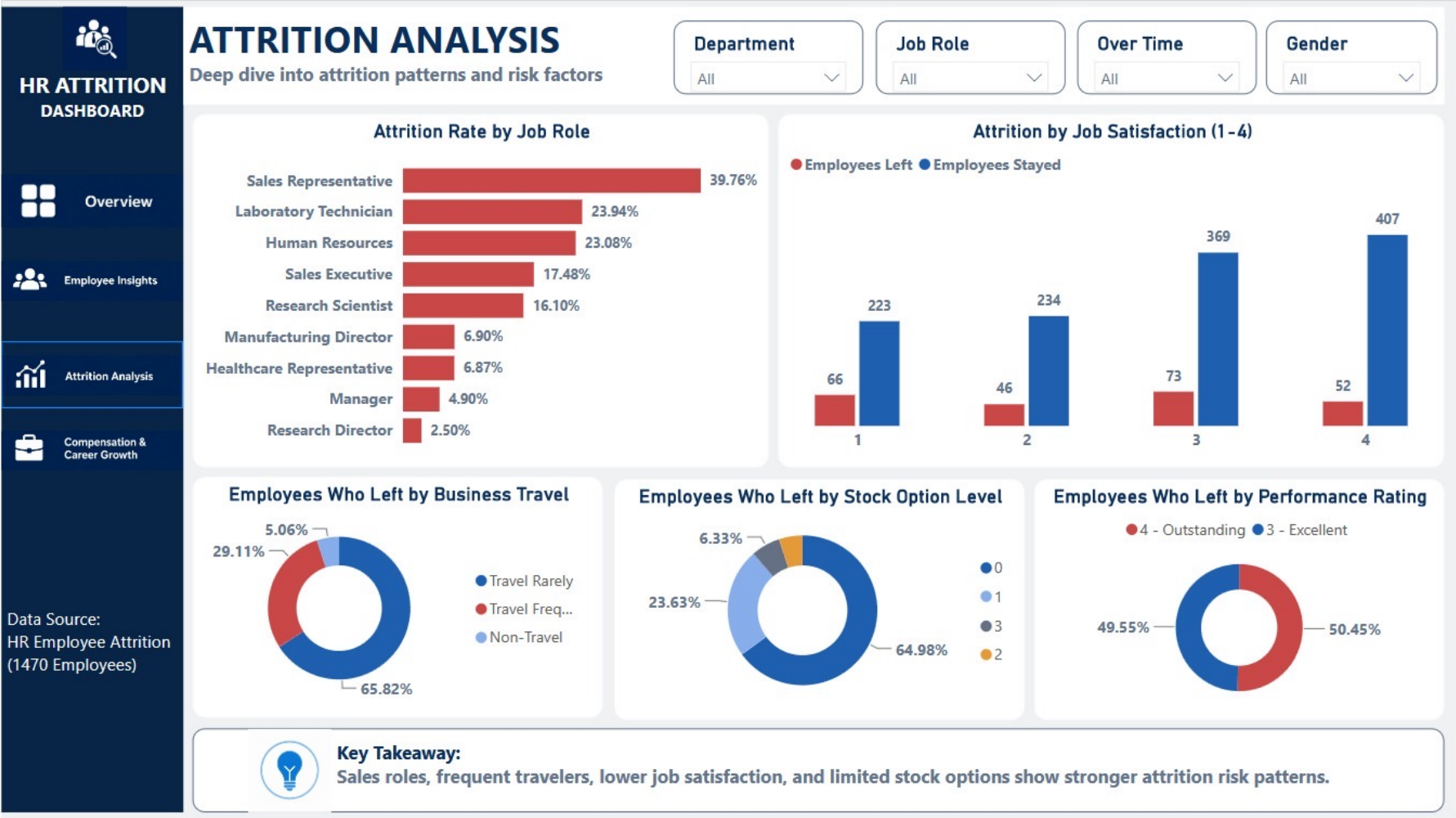


Figure 3. Attrition Analysis dashboard: job-role risk, job satisfaction, travel, stock options, and performance rating.

Power BI Dashboard 4 - Compensation and Career Growth

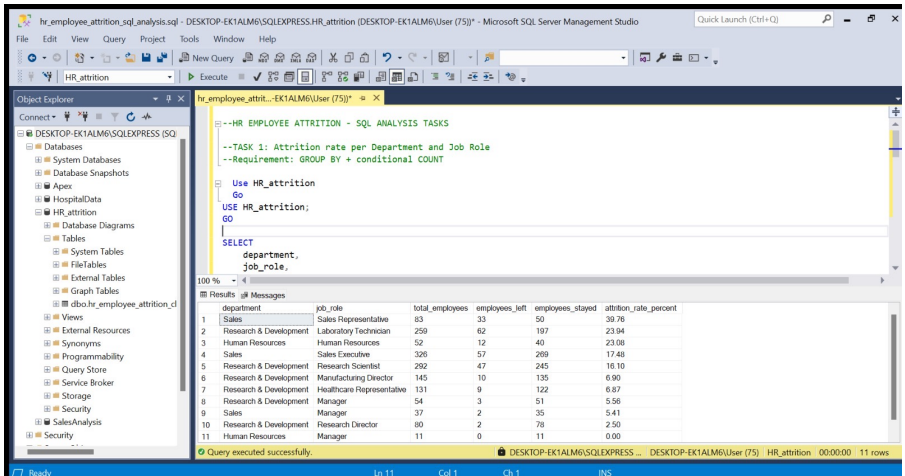
The Compensation and Career Growth page connects job role and job level with average income, attrition by job level, promotion wait time, training exposure, and available salary-hike bands.



Figure 4. Compensation and Career Growth dashboard: pay, job level, promotion wait, training, and salary-hike patterns.

SQL Analysis Evidence and Findings

The SQL phase was performed in SQL Server Management Studio using the cleaned employee table. The queries were designed to validate dashboard insights with transparent, repeatable calculations.



The screenshot shows the SQL Server Management Studio interface. The query editor contains the following SQL code:

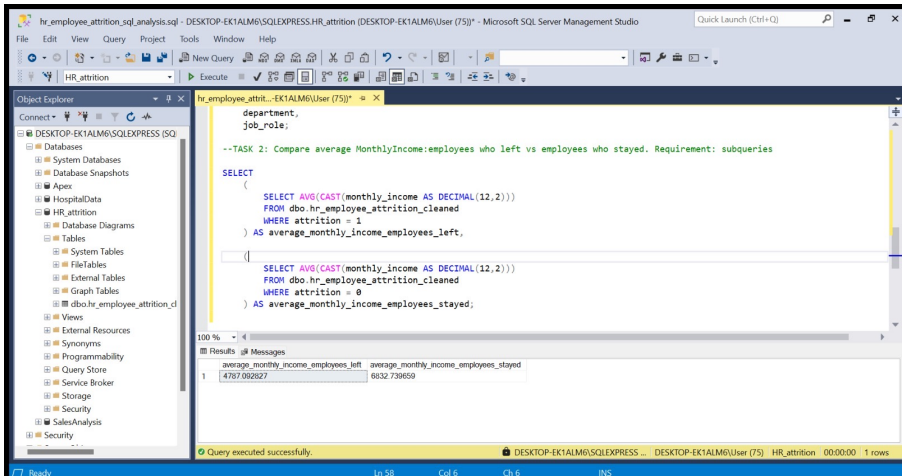
```
--HR EMPLOYEE ATTRITION - SQL ANALYSIS TASKS
--TASK 1: Attrition rate per Department and Job Role
--Requirement: GROUP BY + conditional COUNT

USE HR_attrition
GO
USE HR_attrition;
GO
SELECT
    department,
    job_role,
```

The Results pane displays the following data:

department	job_role	total_employees	employees_left	employees_stayed	attrition_rate_percent
1 Sales	Sales Representative	83	33	50	39.76
2 Research & Development	Laboratory Technician	259	62	197	23.94
3 Human Resources	Human Resources	52	12	40	23.08
4 Sales	Sales Executive	328	57	269	17.48
5 Research & Development	Research Scientist	292	47	245	16.10
6 Research & Development	Manufacturing Director	145	10	135	6.90
7 Research & Development	Healthcare Representative	131	9	122	6.87
8 Research & Development	Manager	54	3	51	5.56
9 Sales	Manager	37	2	35	5.41
10 Research & Development	Research Director	80	2	78	2.50
11 Human Resources	Manager	11	0	11	0.00

SQL Evidence 1 - Attrition rate by department and job role using GROUP BY and conditional COUNT.



The screenshot shows the SQL Server Management Studio interface. The query editor contains the following SQL code:

```
department,
job_role;

--TASK 2: Compare average MonthlyIncome:employees who left vs employees who stayed. Requirement: subqueries

SELECT
(
    SELECT AVG(CAST(monthly_income AS DECIMAL(12,2)))
    FROM dbo.hr_employee_attrition_cleaned
    WHERE attrition = 1
) AS average_monthly_income_employees_left,

(
    SELECT AVG(CAST(monthly_income AS DECIMAL(12,2)))
    FROM dbo.hr_employee_attrition_cleaned
    WHERE attrition = 0
) AS average_monthly_income_employees_stayed;
```

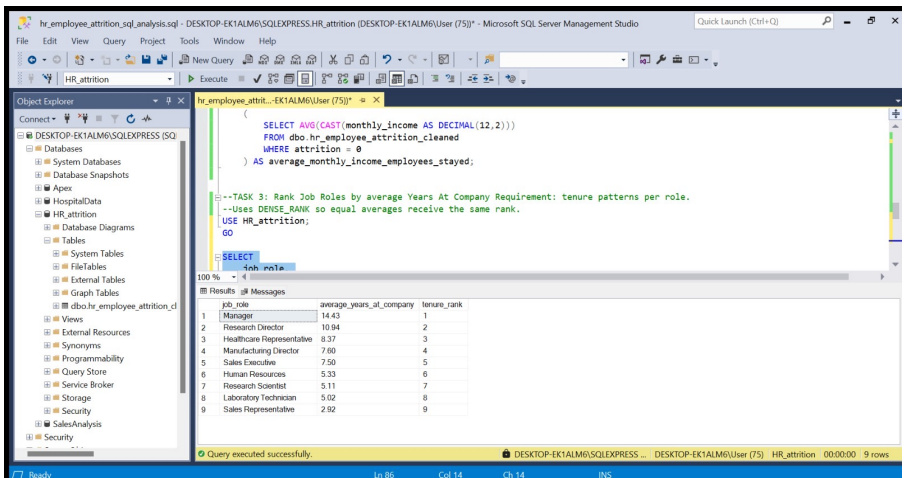
The Results pane displays the following data:

average_monthly_income_employees_left	average_monthly_income_employees_stayed
4767.662627	8632.736659

SQL Evidence 2 - Average monthly income comparison using subqueries.

SQL Evidence Continued - Tenure Ranking

The final SQL task ranks job roles by their average years at the company using DENSE_RANK(). This supports the dashboard finding that roles with the shortest tenure should be examined alongside their attrition levels.



SQL Evidence 3 - Job-role tenure ranking using DENSE_RANK().

Interpretive Note

Managers average 14.43 years at the company, while Sales Representatives average 2.92 years. The contrast is meaningful because Sales Representatives also record the highest attrition rate in the role-level analysis. This provides a clear starting point for targeted retention and career-path review.

SQL Results Interpretation

1. Attrition rate by department and job role

Department	Job Role	Employees	Left	Stayed	Attrition Rate
Sales	Sales Representative	83	33	50	39.76%
Research & Development	Laboratory Technician	259	62	197	23.94%
Human Resources	Human Resources	52	12	40	23.08%
Sales	Sales Executive	326	57	269	17.48%
Research & Development	Research Scientist	292	47	245	16.10%

Sales Representatives are the most exposed group, with 39.76% attrition. Laboratory Technicians and the Human Resources role also exceed the overall organization attrition rate. This supports role-specific retention plans rather than treating attrition as a single organization-wide issue.

2. Average monthly income: employees who left vs stayed



Employees who left had an average monthly income approximately 2,045.65 lower than employees who stayed. This does not establish that compensation caused attrition, because role and job-level composition may differ, but it is a material retention signal that warrants deeper pay-equity and market-benchmark review.

3. Tenure ranking by job role

Rank	Job Role	Average Years at Company
1	Manager	14.43
2	Research Director	10.94
3	Healthcare Representative	8.37
4	Manufacturing Director	7.60
5	Sales Executive	7.50
6	Human Resources	5.33

7	Research Scientist	5.11
8	Laboratory Technician	5.02
9	Sales Representative	2.92

The role with the shortest average tenure is Sales Representative at 2.92 years, which aligns with its highest attrition rate. Managers and Research Directors have the longest tenure, suggesting stronger retention and organizational stability in more senior positions.

Recommendations and Next Steps

1. Build a role-specific retention plan	Start with Sales Representatives, Laboratory Technicians, and Human Resources. Review workload, incentive design, manager support, onboarding quality, career options, and local labor-market pressure.
2. Strengthen the first-year employee experience	The highest tenure-band attrition occurs in the 0-1 year group. Use structured onboarding, 30/60/90-day check-ins, mentoring, and early career conversations.
3. Address high-risk overtime and satisfaction combinations	The dashboard identifies 153 employees who work overtime and report job satisfaction of 1 or 2. Use this as an HR-prioritization list for confidential follow-up and manager coaching.
4. Review pay and progression by role and level	Leavers have lower average income than stayers, while lower job levels show the highest attrition. Investigate base pay, promotion readiness, and internal mobility before adjusting broad policy.
5. Use travel and stock options as targeted discussion areas	The dashboard indicates differences among business-travel and stock-option groups. Segment follow-up by role and department instead of relying on high-level averages.
6. Establish an attrition-monitoring cadence	Refresh the dashboard monthly or quarterly, track action outcomes, and add exit-date, manager, recruitment-cost, and engagement data to improve future analysis.

Conclusion

The HR Employee Attrition analysis shows that retention risk is concentrated in identifiable workforce segments. Sales Representatives have the highest attrition and shortest average tenure; employees who leave have lower average monthly income; early-tenure attrition is elevated; and overtime paired with low job satisfaction creates a practical high-risk signal. The recommended response is targeted: improve role-specific retention, strengthen the first-year experience, review pay and progression by job level, and use the dashboard as a recurring management tool.

Appendix note: dashboard visuals and SQL screenshots are included as project evidence. Percentages and averages are calculated from the cleaned 1,470-record dataset.